

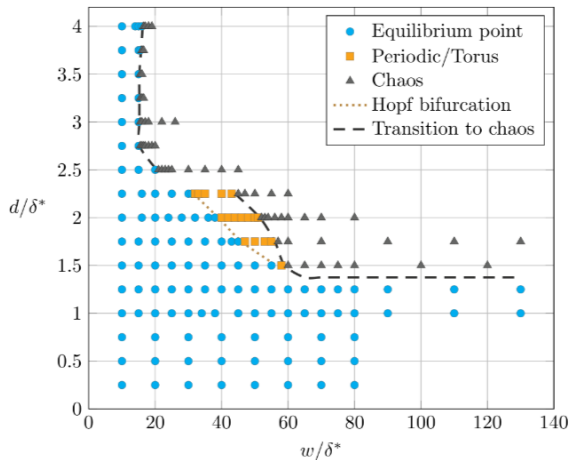
IMPERIAL

Finishing incNS - jeff update

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Stability analysis diagram

- Nonlinear stability results



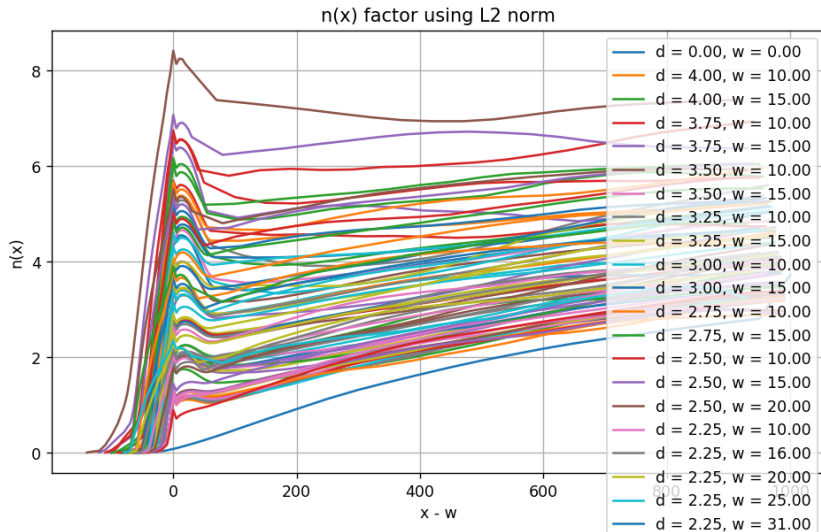
exciting mechanism

- To excite the TS modes, we introduce a divergence-free body force to trigger the instability. The force to the streamfunction is defined as

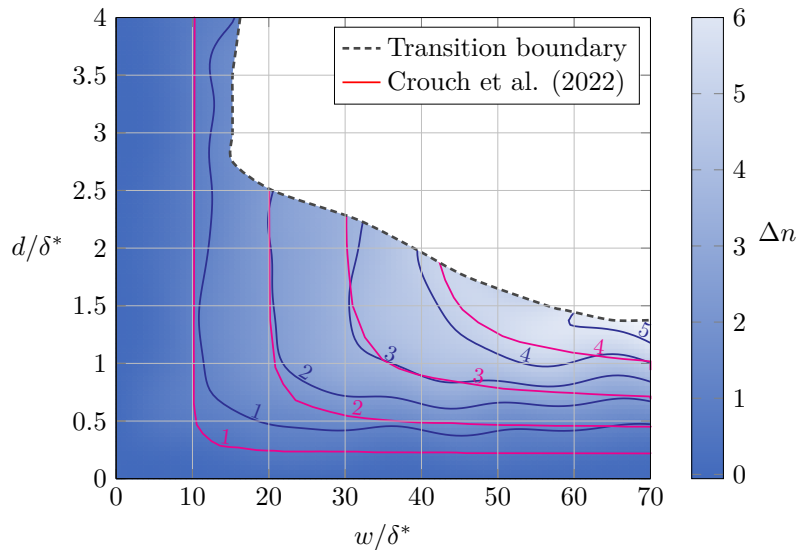
$$\psi = A \exp \left(-w[(x - x_0)^2 + (y - y_0)^2] \right) X_t$$

where X_t is a white gaussian noise. The point (x_0, y_0) is the location of the forcing and it is located in the BL upstream of the gap.

Δn -factors curves



Δn -factors



• $x_G = 450\text{mm}$

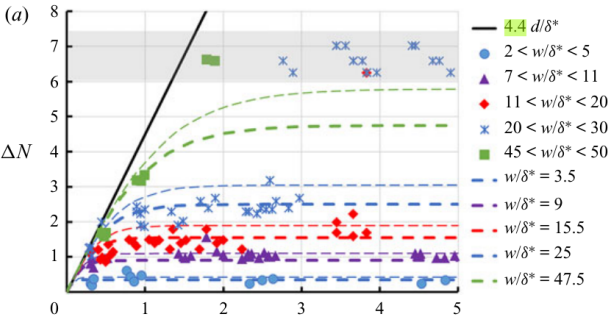


Table 1. Overview of test parameters.

Case	U (m s ⁻¹)	Re_U (m ⁻¹)	δ^* (mm) (at $x_G = 127$ mm)	δ^* (mm) (at $x_G = 450$ mm)
U18	18.3	1.19×10^6	0.52	1.12
U20	20.6	1.33×10^6	0.50	1.05
U22	22.6	1.46×10^6	0.47	1.00
U27	27.5	1.80×10^6	0.43	0.90

